

National Exam of Higher National Diploma-New program – 2020 Session

Spécialty/option : Software Engineering

Paper : Computer Technology

Duration : 4 hours



Credit : 6

SECTION A: MCQ: 40 MARKS

Each question carries 1 mark

1. **A computer Network is _____**
A) Collection of hardware components and computers
B) Interconnected by communication channels
C) Sharing of resources and information
D) All of the Above
2. **A protocol is a(n) _____**
A) Agreement on how communication components and DTE's are to communicate
B) Logical communication channels for transferring data
C) Physical communication channels used for transferring data
D) None of above
3. **Two devices are in network if _____**
A) a process in one device is able to exchange information with a process in another device
B) a process is running on both devices
C) PIDs of the processes running on different devices are same
D) none of the mentioned
4. **What is a Firewall in a Computer Network?**
A) The physical boundary of Network
B) An operating System of Computer Network
C) A system designed to prevent unauthorized access
D) A web browsing Software
5. **Which data communication method is used to transmit the data over a serial communication link?**
A) Simplex
B) Half-duplex
C) Full duplex
D) All of above
6. **Each IP packet must contain _____**
A) Only Source address
B) Only Destination address
C) Source and Destination address
D) Source or Destination address
7. **Software that defines a database, stores the data, supports a query language, produces reports and creates data entry screens is a _____**
A) data dictionary
B) database management system (DBMS)
C) decision support system
D) relational database

8. The database design that consists of multiple tables that are linked together through matching data stored in each table is called a _____.
 A) Hierarchical database B) Network database
 C) Object oriented database D) Relational database
9. Which of the following items is not the advantage of a DBMS?
 A) Improved ability to enforce standards B) Improved data consistency
 C) Local control over the data D) Minimal data redundancy
10. Two different terms are used to describe the characteristics of interest for an entity. They are attributes and _____.
 A) classes B) entities C) properties D) traits
11. Which of the below given sorting techniques has highest best-case runtime complexity.
 A) quick sort B) selection sort C) insertion sort D) bubble sort
12. Which of the following declaration is illegal?
 A) char *str = "Best C programming classes by Joe Digital Academy";
 B) char str[] = "Best C programming classes by Joe Digital Academy";
 C) char str[20] = "Best C programming classes by Joe Digital Academy";
 D) char[] str = "Best C programming classes by Joe Digital Academy";
13. Can a linear search recursive algorithm and binary search recursive algorithm be performed on an unordered list?
 A) Binary search can't be used B) Linear search can't be used
 C) Both cannot be used D) Both can be used
14. The array is as follows: 1,2,3,6,8,10. Given that the number 17 is to be searched. At which call it tells that there's no such element? (By using linear search(recursive) algorithm).
 A) 7th call B) 9th call
 C) 17th call D) The function calls itself infinite number of times
15. What is the header file for the string class?
 A) #include<ios> B) #include<str>
 C) #include<string> D) #include<stio>
16. Which of the following displays the unique values of a column?
 SELECT _____ dept_name
 FROM instructor;
 A) All B) From C) Distinct D) Name
17. Which of the following is a correct way to declare a multidimensional array in Java?
 A) int [][]arr; B) int arr[[]]; C) int arr [][]; D) int[[]] arr;
18. The UML class diagram also referred to as _____.
 A) dynamic modeling. B) static modeling.
 C) object modeling. D) test modeling.
19. An/A _____ is a user playing a role with respect to the system.
 A) developer. B) enduser. C) customer. D) actor.
20. _____ software is expected to provide a solution to a problem.
 A) System. B) Application. C) Engineering. D) Design.
21. _____ are an important mechanism for classifying objects.
 A) Methods. B) Objects. C) Classes. D) Code.
22. _____ diagrams are used to illustrate data structures, and the static snapshots instances of the things found in the class diagrams.

- A) use case B) Object C) Collaboration D) Sequence
23. A function in JavaScript with no return value is called
A) Procedures B) Method C) Static function D) Dynamic function
24. The <a> and tags in HTML are use for _____
A) Adding images B) Adding Text C) Adding links D) Adding headings
25. Which of the following HTML tag is use to make text italic?
A) <i> B) <italic> C) <ij> D) <it>
26. What tag is used to display a picture in a HTML page?
A) <picture> B) <image> C) D) <src>
27. Devices on one network can communicate with devices on another network via a _____
A) File Server B) Utility Server C) Printer Server D) Gateway
28. What is the output of this program in C?

```
int main(void)
{
    int i = 3;
    for (;i++=0;) printf("--%d\\n,i);
}
```

A) A character B) A set of character C) 4 D) Compiler Error
29. What is the most restrictive access modifier?
A) Private B) Public C) Protected D) Restricted
30. Which of the following loops will execute at least once?
A) For B) Do-while C) If D) While
31. PHP File have a default file extension of _____
A) .html B) .xml C) .php D).pdf
32. In a Hierarchical model records are organized as _____
A) Graph. B) List. C) Links. D) Tree.
33. In an E-R diagram attributes are represented by _____
A) rectangle. B) square. C) ellipse. D) triangle.
34. The language used in application programs to request data from the DBMS is referred to as the _____
A) DML B) DDL C) VDL D) SDL
35. In the _____ traversal we process all of a vertex's descendants before we move to an adjacent vertex.
A) Depth First B) Breadth First C) With First D) Depth Limited
36. Inserting an item into the stack when stack is not full is called _____ Operation and deletion of item form the stack, when stack is not empty is called _____ operation.
A) push, pop B) pop, push C) insert, delete D) delete, insert
37. _____ is a pile in which items are added at one end and removed from the other.
A) Stack B) Queue C) List D) None of the above
38. The _____ multithreading model multiplexes many user-level threads to a smaller or equal number of kernel threads.
A) many-to-one model B) one-to-one model
C) many-to-many model D) many-to-some model
39. _____ is the number of processes that are completed per time unit.
A) CPU utilization B) Response time
C) Turnaround time D)Throughput
40. Which of the following devices forwards data packets to all connected ports?
A) Router B) Switch C) Bridge D) Hub

SECTION B : 60 MARKS

Question I Programming (20 marks)

1. Give the full mean and the function of the following:

- PHP
- JAVASCRIPT
- SQL
- HTML (1 x 4 marks)

(Note: For questions 2 and 3 below, you may use any one of the following programming languages: C, C++, JAVA, PHP, or Visual Basic)

2. a) What is a recursive function? (1 mark)
b) Define a function called *factorial* that accepts one parameter and returns the computed value. (3 marks)
c) Write a program that does the following:
- prompts a user for a non-negative,
- calls the factorial function, and
- prints the factorial of the number entered by the user. (7 marks)
3. Write a C Program that read an integer number in Decimal and converts it into Binary Number System. (5 marks)

Question II: Databases (15 marks)

1. Write down the SQL statement to create a database called **myexamdb**. (1 mark)
2. Write the SQL statement to drop a database called **myexamdb**. (1 mark)
3. Can we have two tables with the same name in mysql? (2marks)
4. Give 3 examples of Database Management System. (2marks)

Consider the following from question 5 to 6

Customer(CustID: int(10), CustName: varchar(50), Address: varchar(50), CustPhone: varchar(50), town: varchar(50), age int(10), gender: (character)

5. Write the sql statement to create the table customer. (2 marks)
6. Provide the SQL statement to delete all male customers between 20 and 40. (2marks)
7. SQL statement to list all the customer living in "Douala" who have a name which starts with "sh". (3 marks)
8. Write the SQL statement to display the number of customers per town. (2marks)

Question III: NETWORKING (15 Marks)

As far as the client/server networking model is concerned, answer the following questions:

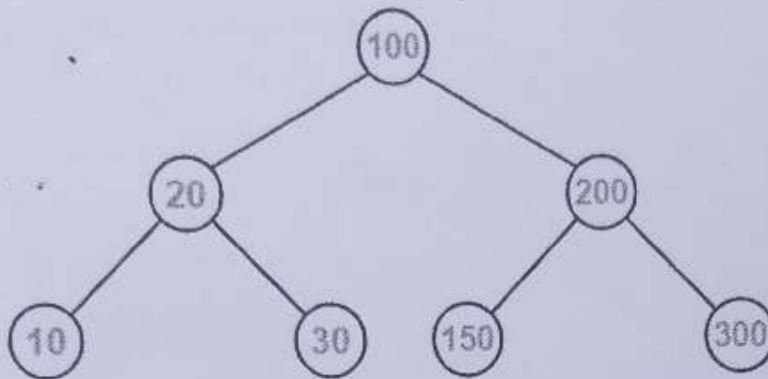
1. What is a server? What is a client? Give one example of each. (4marks)
2. Name three protocols regulating the communication between clients and servers. (3marks)
3. Name three services that can be provided by network servers. (3marks)
4. Is there a direct connection between clients in this model? If not, how does a client proceed to get connected to another client? (4marks)
5. Where did TCP/IP get its name from? (1marks)

Question IV: DATA STRUCTURE AND ALGORITHM (10marks)

Traversal is a process to visit all the nodes of a tree and may print their values too. Because, all nodes are connected via edge links we always start from the root head node. That is, we cannot random access a node in tree. There are three ways which we use to traverse a tree –

- In-order Traversal
- Pre-order Traversal
- Post-order Traversal

1. State the algorithms of the tree traversals mentioned above. (2 x 3 marks)
2. Consider the following binary search tree-



Now, write the traversal sequences for this binary search tree in;

- a) In-order Traversal
- b) Pre-order Traversal
- c) Post-order Traversal. (1 x 3 marks)

3. What is a data structure? Name an example of a linear data structure. (1 mark)